Credit Name: CSE 2140 2nd Language Programming

Assignment Name: Grade.java

The following questions can help you in thinking critically about your problem-solving processes:

Understanding the Problem

How did you approach understanding the challenge?

i just looked at how normal grading scales work where different percentage brackets give you a specific letter grade.

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

Making sure the number cutoffs didn't overlap or miss any numbers was a bit tricky. I fixed it by using clear limits like >= 80 && <= 89 for each grade bracket.

Planning the Solution

Did you create a plan or break the problem into smaller steps before coding?

Yeah, I planned to first grab the user input with a Scanner, then run through the if-else conditions, and finally print the letter grade out.

How did you decide on the tools, data structures, or algorithms to use?

I used an int for the percentage since grades are whole numbers, and a Scanner to read what gets typed in.

Implementation

Did you write the code in small pieces or attempt the entire solution at once?

I did it piece by piece, set up the input first, added the if-else ladder, and then tested it

How did you test your solution along the way to make sure it was working?

I tested boundary numbers like 90, 89, and failing scores under 60 to see if every part of the if-else chain actually worked.

Overcoming Challenges

What part of the problem was the most difficult for you?

At first, the program would ask for the input and then hang without showing the grade. I fixed that by separating the prompt from the output print statement so the console flushed the text properly.

How did you handle moments when you felt stuck or unsure of what to do next?

I just manually walked through the code with a fake number to see which if branch it would drop into.

Learning

Was there anything you learned that you think will help you with future challenges?

Getting used to writing multi-way if-else if blocks will definitely make it easier to handle tiered menus or grading apps later on.